

Postdoctoral research fellow at the University of Colorado Denver | Anschutz Medical Campus with over 8 years of research experience processing and analyzing neural, imaging, and behavioral data. Trained in basic science research and methodology with a passion in designing and building devices and distilling large biomedical and neural data sets.

## SKILLS

|                       |                                                                                                                                                                                                             |
|-----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Tools and Languages   | Matlab, Python (NumPy, SciPy, matplotlib, pandas), Git, L <sup>A</sup> T <sub>E</sub> X, HPC scheduler (SLURM), SolidWorks, Autodesk Inventor, Microsoft Office Suite, Adobe Illustrator, Affinity Designer |
| Quantitative Research | Research Design, Research Dissemination, Project Management, Interpersonal and Collaborative Communication, Data Analysis and Visualization                                                                 |

## EDUCATION

**Phd, Biomedical Engineering**, Carnegie Mellon University Aug 2016 — Sep 2022  
*Thesis: Dissection of detailed motor behaviors and circuit functions of the basal ganglia in health and disease*

**BS, Bioengineering**, George Mason University Aug 2010 — Jun 2014

## RESEARCH AND WORK EXPERIENCE

**POSTDOCTORAL FELLOW** Nov 2022 — Present  
University of Colorado  
Advisor — Abigail Person, PhD  
Aurora, CO

- Study kinematic adjustments during a skilled forelimb reaching tasks in rodents with simultaneous recordings and perturbations to cerebellar and basal ganglia circuits.
- Build data pipelines to acquire videos and synchronize with 2-photon calcium imaging data during behavior.
- Present research regularly at international conferences and internal seminars.

**GRADUATE STUDENT RESEARCHER** Aug 2016 — Oct 2022  
Carnegie Mellon University  
Advisors — Aryn Gittis, PhD and Steven Chase, PhD  
Pittsburgh, PA

- Designed and constructed novel behavioral devices to study kinematic adjustments during motor learning tasks in rodents.
- Built data pipelines to acquire high frame rate video files (100s of GBs) and analyze data in MATLAB and Python using regression, time series analysis, and probabilistic frameworks.
- Mentored students (2 undergraduates, 2 masters) and collaborated with lab mates to apply my computational skills to their projects.

**POSTBACCALAUREATE IRTA FELLOW** Jul 2014 — Aug 2016  
National Institutes of Health (NIDDK)  
Advisor — Alexxai Kravitz, PhD  
Bethesda, MD

- Constructed a low-cost, home cage compatible automatic pellet dispensing device to obtain high temporal resolution data for feeding behavior and patterns. [web]

**UNDERGRADUATE STUDENT RESEARCHER** Apr 2013 — Jul 2014  
George Mason University  
Advisor — Wilsaan Joiner, PhD  
Fairfax, VA

- Designed and performed psychophysical studies on human subjects to study the retention of motor adaptation upon different methods of applied perturbing force.
- Analyzed data sets (10s of GBs) in MATLAB to identify changes in reaching movements with motor adaptation.

## LEADERSHIP AND TEAM EXPERIENCE

**PYTHON TEACHING ASSISTANT** Sep 2023  
Simons Foundation  
New York, NY

- Served as a coding teaching assistant for an 8-week undergraduate research fellowship (SURFiN), offering coding support, holding office hours, and contributing to course material development.

**BOOTCAMP TEACHING ASSISTANT** Aug 2021  
Carnegie Mellon University  
Pittsburgh, PA

- Developed an intensive 3-day "Computational Neuroscience Bootcamp" for incoming graduate students.
- Guided students in the acquisition and analysis of a sample data set to develop a broad foundation of computational tools.